

The empirical recurrence for OEIS A296383: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A296383 records “Empirical recurrence of order 92”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 142$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 92, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 92 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A296383 is “Number of $n \times 5$ 0..1 arrays with each 1 horizontally, vertically or antidiagonally adjacent to 1 or 2 neighboring 1s.”. It has offset 1 and begins

12, 279, 3237, 46786, 684894, 9703136, 138882856, 1988460073, ...

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (Dec 11 2017 11:28:47, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 142$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 142$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 284$ exact terms of the model returns order 92 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	13	c_{24}	294945083471	c_{47}	−380681062561	c_{70}	−640331305
c_2	2	c_{25}	−196932969804	c_{48}	383546532486	c_{71}	863335483
c_3	373	c_{26}	−534710334912	c_{49}	256892964311	c_{72}	−577959611
c_4	−2138	c_{27}	803372554765	c_{50}	−420717798517	c_{73}	230727507
c_5	5098	c_{28}	447323897519	c_{51}	79493471143	c_{74}	−44083369
c_6	−42895	c_{29}	−1690570185629	c_{52}	53849563691	c_{75}	175467
c_7	235088	c_{30}	537488492231	c_{53}	172716145445	c_{76}	−10279142
c_8	−768321	c_{31}	1872487895980	c_{54}	−292496214291	c_{77}	20083161
c_9	3250494	c_{32}	−2073215357463	c_{55}	119397343401	c_{78}	−18177260
c_{10}	−14955502	c_{33}	−260580833261	c_{56}	41883787203	c_{79}	11050575
c_{11}	50745217	c_{34}	2141369080606	c_{57}	31623981049	c_{80}	−5105507
c_{12}	−140623935	c_{35}	−1775777230612	c_{58}	−191785577099	c_{81}	2056018
c_{13}	337049428	c_{36}	−20380000553	c_{59}	216050945684	c_{82}	−936693
c_{14}	−511830200	c_{37}	1621475537482	c_{60}	−102069287620	c_{83}	557944
c_{15}	−4758328	c_{38}	−2027664463370	c_{61}	−6135324079	c_{84}	−354716
c_{16}	1521676378	c_{39}	1275215450963	c_{62}	27912135385	c_{85}	198838
c_{17}	−2318843717	c_{40}	102461589246	c_{63}	3391785764	c_{86}	−92371
c_{18}	2541630826	c_{41}	−1492440219474	c_{64}	−28264012818	c_{87}	35006
c_{19}	−8032781031	c_{42}	1883447837322	c_{65}	27268973755	c_{88}	−10705
c_{20}	8268667888	c_{43}	−587328347441	c_{66}	−13917073027	c_{89}	2591
c_{21}	36080393031	c_{44}	−1339039295909	c_{67}	3367023221	c_{90}	−475
c_{22}	−91622345481	c_{45}	1925600884969	c_{68}	389653263	c_{91}	60
c_{23}	−21179091603	c_{46}	−813129570723	c_{69}	−209524280	c_{92}	−4

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 92, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $92 + 4$ terms removed returns order 92 as well, so $\deg q_{\min} = 92 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $92 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 92 that this sequence satisfies — and that family turns out to have exactly one member.

References

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